

Space Exploration: Past, Present, and Future

Space exploration has captivated humanity since the launch of Sputnik 1 by the Soviet Union on 4 October 1957, marking the beginning of the Space Age. Since then, humans have landed on the Moon, deployed robotic probes to every planet in the solar system, and established a continuous human presence in low Earth orbit.

1. The Space Race

The Cold War rivalry between the United States and the Soviet Union drove rapid advances in rocket technology. Key milestones include Yuri Gagarin's orbital flight (1961), the first spacewalk by Alexei Leonov (1965), and NASA's Apollo 11 mission that put Neil Armstrong and Buzz Aldrin on the Moon in July 1969.

2. The International Space Station

The ISS is a collaborative project involving NASA, Roscosmos, ESA, JAXA, and CSA. It has been continuously inhabited since November 2000 and serves as a microgravity laboratory for research in biology, physics, astronomy, and medicine.

3. Mars Exploration

NASA's Perseverance rover, which landed in Jezero Crater in February 2021, is searching for signs of ancient microbial life and collecting rock samples for future return to Earth. China's Tianwen-1 mission also successfully deployed the Zhurong rover in May 2021.

4. Artemis and Lunar Return

NASA's Artemis programme aims to return humans to the Moon, including the first woman and first person of colour, using the Space Launch System (SLS) and the Orion capsule. The Lunar Gateway, a small space station in lunar orbit, will support long-duration surface missions and serve as a staging point for Mars.

5. Commercial Spaceflight

Companies such as SpaceX, Blue Origin, and Rocket Lab have transformed the economics of launch. SpaceX's reusable Falcon 9 has dramatically cut costs, and Starship—the largest rocket ever built—is intended to carry humans to Mars. Commercial crew programmes have ended US dependence on Russian Soyuz for ISS transport.